Set Theory Primer

Will use set to define concept of prob. space.

Def: Set : collection of well-defined objects (called element)

$x \in A$: x belongs to set A.

$A = \{1, 2, 3\}$; $B = \{x \in \mathbb{N} : x \text{ even}\}$.

Def: Set A is subset of set B if every element in A is also an element of B. $A \subseteq B$.

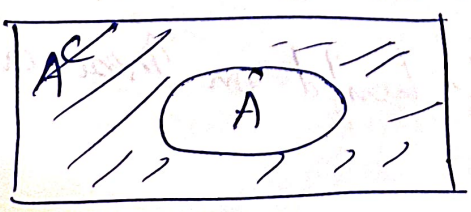
Def: Two sets A, B are equal if $A \subseteq B$ & $B \subseteq A$.

Two special sets: Universe $\mathcal{U}$ $A \subseteq \mathcal{U}$ $\forall A$

Null set $\emptyset$ $\emptyset \subseteq A$ $\forall A$

Operations on sets ① Complement

For $A \subseteq \mathcal{U}$, $A^c = \{x \in \mathcal{U} \mid x \notin A\}$



Union & Intersection - $A_1 \cup A_2 = \{x \mid x \in A_1 \text{ or } x \in A_2\}$

$A_1 \cap A_2 = \{x \mid x \in A_1 \text{ and } x \in A_2\}$

Can extend this to any finite collection or even infinite collection which is indexed

Properties - ① $\left(\bigcap_{i\in I} A_i\right) \cup B = \bigcap_{i\in I} (A_i \cup B)$

② $\left(\bigcup_{i\in I} A_i\right) \cap B = \bigcup_{i\in I} (A_i \cap B)$.

③ $\left(\bigcup_{i\in I} A_i\right)^c = \bigcap_{i\in I} A_i^c$

④ $\left(\bigcap_{i\in I} A_i\right)^c = \bigcup_{i\in I} A_i^c$

} De Morgan's Laws.

Pf of ① - show $\underbrace{\left(\bigcap_{i\in I} A_i\right) \cup B}_{Z} \subseteq \underbrace{\bigcap_{i\in I} (A_i \cup B)}_{Y}$

Say $x \in Z \Rightarrow x \in \bigcap_{i\in I} A_i$ or $x \in B$.

Say $x \in B \Rightarrow x \in A_i \cup B \ \forall i \Rightarrow x \in Y$

Say $x \notin B \Rightarrow x \in A_i \ \forall i \Rightarrow x \in Y$

Now show $\underbrace{\bigcap_{i\in I} A_i \cup B}_{Y} \subseteq \underbrace{\left(\bigcap_{i\in I} A_i\right) \cup B}_{Z}$

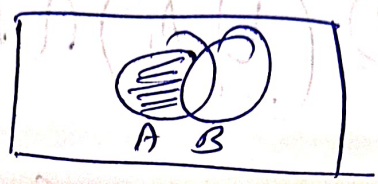
Say $x \in Y \Rightarrow x \in A_i \cup B \ \forall i$

Say $x \in B \Rightarrow x \in Z$

Say $x \notin B \Rightarrow x \in A_i \ \forall i \Rightarrow x \in \bigcap_i A_i \Rightarrow x \in Z$.

Relative complement (set difference)

$$A \setminus B = \{x \in \mathcal{U} \mid x \in A \ \& \ x \notin B\}$$



Cartesian product: $A \times B = \{(x, y) \mid x \in A, y \in B\}$

Extend to n sets $A_1, A_2 \ldots A_n$.

If A_i's are the same, denote it as A^n.

$$\mathbb{R}^n = \{(x, y) \mid x, y \in \mathbb{R}\}$$

Power set – 2^A is the set of all subsets of A

$$A = \{1, 2\} \qquad 2^A = \{\phi, \{1\}, \{2\}, \{1, 2\}\}$$

If A has n elements $\Rightarrow$ 2^A has 2^n elements

Probability space 31/7

Random experiment. Outcome ω.

① Sample space Ω: set of all possible outcomes

Eg. $\Omega = \{H, T\}$ in toss of a coin – finite

$\Omega = \mathbb{N}$ if # tosses before head. – Countably infinite

$\Omega = \mathbb{R}_+$ if tracking speed of hitting ground – uncountably infinite

② Events – Subset of Ω to which we will assign prob.

In a random experiment, we will say event $\mathcal{E}$ occured if the outcome $\omega \in \mathcal{E}$.

Eg Say toss coins 3 times. $\Omega = \{H, T\}^3$

Events → No head → $\mathcal{E} = \{\{T, T, T\}\}$

First toss tail → $\mathcal{E} = \{\{T, H, H\}, \{T, H, T\}, \{T, T, H\}, \{T, T, T\}\}$

Will want some properties in collection of events.

① If A is event, A^c is event

② A, B events then $A \cup B$ is event

③ Ω is event.

$(\Omega, \mathcal{F}, \mathbb{P})$ → Prob. space

Ω → sample space, $\mathcal{F}$ → σ-algebra on Ω : defines the events.

$\mathbb{P}$ → prob. measure : defines prob. of each event

Def → A prob. measure $\mathbb{P}$ on $(\Omega, \mathcal{F})$ is a fn $\mathbb{P}: \mathcal{F} \to [0,1]$ s.t

(i) $\mathbb{P}(\phi) = 0$ (ii) $\mathbb{P}(\Omega) = 1$, (iii) Given seq. of disjoint sets $\{A_i, i \geq 1\}$ in $\mathcal{F}$, $\mathbb{P}\left(\bigcup_{i \geq 1} A_i\right) = \sum_{i \geq 1} \mathbb{P}(A_i)$

Eg → Flip coin once $\Omega = \{H, T\}$

$\mathcal{F} = 2^{\Omega} = \{\phi, \{H\}, \{T\}, \{H, T\}\}$

$\mathbb{P}(\phi) = 0$, $\mathbb{P}(\{H, T\}) = 1$. If $\mathbb{P}(\{H\}) = p$, then $\mathbb{P}(\{T\}) = 1 - p$.

Properties of Prob. space $(\Omega, \mathcal{F}, \mathbb{P})$

(1) $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$ [$A \cup A^c = \Omega$]

(2) Event A, B, $A \subseteq B$, $\mathbb{P}(A) \leq \mathbb{P}(B)$ [$A \cup \underbrace{B \setminus A}_{\in \mathcal{F}} = B$]

(3) A, B $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$
[$A \cup B = A \cup (B \setminus A)$, $B = (B \setminus A) \cup (A \cap B)$]

(4) For n events $A_1, A_2 \ldots A_n$

$$\mathbb{P}\left(\bigcup_{i=1}^{n} A_i\right) = \sum_i \mathbb{P}(A_i) - \sum_{i<j} \mathbb{P}(A_i \cap A_j) + \sum_{i<j<k} \mathbb{P}(A_i \cap A_j \cap A_k) + \cdots + (-1)^n \mathbb{P}(A_1 \cap A_2 \cap A_3 \ldots \cap A_n)$$

[Inclusion-Exclusion] [Induction]

(5) Union bound - $\mathbb{P}\left(\bigcup_{i\in I} A_i\right) \le \sum_{i\in I} P(A_i)$

I: Countable set

Pf - Express $\bigcup_{i\in I} A_i$ as union of disjoint sets

$B_1 = A_1$, $B_2 = A_2 \setminus A_1$ ... $B_n = A_n \setminus \bigcup_{i=1}^{n-1} A_i$ ...

B_i's are disjoint, $B_i \subseteq A_i$ $\bigcup_{i\in I} B_i = \bigcup_{i\in I} A_i$

$$\mathbb{P}\left(\bigcup_{i\in I} A_i\right) = \mathbb{P}\left(\bigcup_{i\in I} B_i\right) = \sum_{i\in I} \mathbb{P}(B_i) \le \sum_{i\in I} \mathbb{P}(A_i)$$

Generalization of Union bound : Bonferroni Ineq

$$\mathbb{P}\left(\bigcup_{i\in I} A_i\right) \ge \sum \mathbb{P}(A_i) - \sum_{i<j} \mathbb{P}(A_i \cap A_j)$$

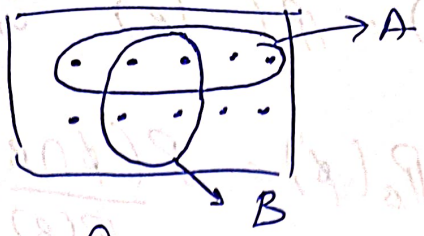
$$\mathbb{P}\left(\bigcup_{i\in I} A_i\right) \le \sum \mathbb{P}(A_i) - \sum_{i<j} \mathbb{P}(A_i \cap A_j) + \sum_{i<j<k} \mathbb{P}(A_i \cap A_j \cap A_k)$$

Conditional Probability

$\Omega \Rightarrow$ Contains 10 pts $\quad n = 10$

Random exp. picks one pt uniformly. $P(\{i\}) = \frac{1}{10} \ \forall i$

For event E, $P(E) = \frac{|E|}{10}$



$P(A) = \frac{5}{10}$

$P(B) = \frac{4}{10}$

If B occured, then each of these 4 pts is equally likely. Out of these, 2 are the ones relevant to A

If B occured, what is the prob. of A occuring

$P(A|B)$ intuitively should be $\frac{N(A \cap B)}{N(B)}$

$$= \frac{N(A \cap B)/n}{N(B)/n} = \frac{P(A \cap B)}{P(B)}$$

Defn → Given $(\Omega, \mathcal{F}, P)$, events $A, B \in \mathcal{F}$ s.t $P(B) > 0$

define $P(A|B) \triangleq \frac{P(A \cap B)}{P(B)}$

Eg 2 coins C_1 → fair, C_2 → always shows heads.

Select one & toss.

$\Omega = \{C_1, C_2\} \times \{H, T\}$

$= \{\{C_1, H\}, \{C_1, T\}, \{C_2, H\}, \{C_2, T\}\}$

$\quad \downarrow 1/4 \quad \downarrow 1/4 \quad \downarrow 1/2 \quad \downarrow 0$

$E \rightarrow$ observed tails, $A_1 \rightarrow$ picked C_1, $A_2 \rightarrow$ picked C_2

$E = \{\{C_1, T\}, \{C_2, T\}\}$ $\quad A_1 = \{\{C_1, H\}, \{C_1, T\}\}$ $\quad A_2 = \{\{C_2, T\}, \{C_2, H\}\}$

$P(E) = 1/4$ $\quad P(A_1) = 1/2$ $\quad P(A_2) = 1/2$

$P(\text{tails given we toss Coin 1}) = \frac{P(E \cap A_1)}{P(A_1)} = \frac{P(\{C_1, T\})}{1/2} = \frac{1/4}{1/2} = \frac{1}{2}$

$P(\text{tails given we toss coin 2}) = \frac{P(\{C_2, T\})}{P(A_2)} = 0$

Claim - Given $(\Omega, \mathcal{F}, P)$ & event B s.t $P(B) > 0$.

$P_B(\cdot) = P(\cdot|B)$ is a valid prob. measure.

Pf → ① $P_B(\phi) = \frac{P(\phi \cap B)}{P(B)} = 0$

② $P_B\left(\bigcup_{i \in I} A_i\right) = \frac{P\left(\left[\bigcup_{i \in I} A_i\right] \cap B\right)}{P(B)} = \frac{P(\bigcup (A_i \cap B))}{P(B)}$

(A_i are disjoint)

$= \sum \frac{P(A_i \cap B)}{P(B)} = \sum P_B(A_i)$

③ $P_B(\Omega) = \frac{P(\Omega \cap B)}{P(B)} = 1$

Key properties ① Multiplication / chain Rule

② Law of Total prob.

③ Bayes Rule

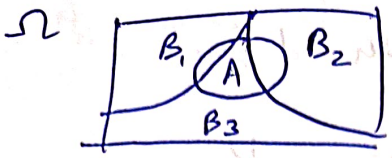
① $P(A \cap B) = P(A) \cdot P(B|A) = P(B) \cdot P(A|B)$

$P(A \cap B \cap C) = \frac{P(A) \cdot P(A \cap B \cap C)}{P(A)} = \frac{P(A) \cdot P(\tilde{E}) \cdot P(C|\tilde{E})}{P(A)}$

$= P(A) \cdot \frac{P(A \cap B)}{P(A)} \cdot P(C|A \cap B) = P(A) \cdot P(B|A) \cdot P(C|A \cap B)$

$P\left(\bigcap_{i \in I} A_i\right) = P(A_1) \cdot \prod_{\substack{i \in I \\ i \geq 2}} P(A_i | (A_1 \cap A_2 \cdots \cap A_{i-1}))$

(2) Law of Total Prob.



$A \in \mathcal{F}$, $\{B_i : i \in I\}$ partition Ω.

Assume $\mathbb{P}(B_i) > 0 \ \forall i \in I$.

Then $\mathbb{P}(A) = \sum_{i \in I} \mathbb{P}(B_i) \cdot \mathbb{P}(A|B_i)$

(Convex combination of $\mathbb{P}(A|B_i)$)

Pf: $= \sum_{i \in I} \mathbb{P}(A \cap B_i) = \mathbb{P}\left(\bigcup_{i \in I} (A \cap B_i)\right)$

$= \mathbb{P}(A)$.

Eg – Two coins eg. $\mathcal{E} \to$ obs tails

$A_1 \to$ pick coin 1

$A_2 \to$ pick coin 2

A_1, A_2 partition underlying Ω

$$\mathbb{P}(\mathcal{E}) = \mathbb{P}(A_1) \cdot \mathbb{P}(\mathcal{E}|A_1) + \mathbb{P}(A_2) \cdot \mathbb{P}(\mathcal{E}|A_2)$$

$$= \frac{1}{2} \times \frac{1}{2} + \frac{1}{2} \times 0 = \frac{1}{4}.$$

(3) Bayes Rule – Forms basis of Bayesian inference in ML

Start with prior belief & update it based on data

Given – A, $\{B_i\}$ that partition Ω $\quad \mathbb{P}(B_i)$ = initial belief that B_i occurs.

$\mathbb{P}(B_i|A)$ = updated belief.

$$\mathbb{P}(B_i|A) = \frac{\mathbb{P}(B_i \cap A)}{\mathbb{P}(A)} = \frac{\mathbb{P}(B_i) \cdot \mathbb{P}(A|B_i)}{\sum_{i \in I} \mathbb{P}(B_i) \cdot \mathbb{P}(A|B_i)}$$

Eg 2 coins $P(A_1) = \frac{1}{2}$, $P(A_2) = \frac{1}{2}$

Leave this

Suppose $\mathcal{E}$ is true : observed tails

$$P(A_1 | \mathcal{E}) = \frac{P(A_1 \cap \mathcal{E})}{P(\mathcal{E})} = \frac{1/4}{1/4} = 1$$ → Makes sense

$$P(A_2 | \mathcal{E}) = \frac{P(A_2 \cap \mathcal{E})}{P(\mathcal{E})} = \frac{0}{1/4} = 0$$

Suppose $\mathcal{E}^c$ is true

$$P(\mathcal{E}^c) = 1 - P(\mathcal{E}) = 3/4$$

$$P(\mathcal{E}^c) = P(A_1) P(\mathcal{E}^c | A_1) + P(A_2) P(\mathcal{E}^c | A_2)$$

$\frac{1}{2}$ $\frac{1}{2}$

$\frac{1}{2} \times 1 = 3/4$

$$P(A_1 | \mathcal{E}^c) = \frac{P(A_1 \cap \mathcal{E}^c)}{P(\mathcal{E}^c)} = \frac{1/4}{3/4} = \frac{1}{3}$$

$$P(A_2 | \mathcal{E}^c) = \frac{P(A_2 \cap \mathcal{E}^c)}{P(\mathcal{E}^c)} = \frac{1/2}{3/4} = \frac{2}{3}$$

Updated belief

Independence b/w events

$(\Omega, \mathcal{F}, P)$: A, B are said to be ind. if $P(A \cap B) = P(A) \cdot P(B)$

If $P(B) > 0 \Rightarrow P(A|B) = P(A)$

Occurence of B does not alter likelihood of A

Are $\mathcal{E}, A_1$ independent? $P(\mathcal{E} \cap A_1) = 1/4 \neq P(\mathcal{E}) \cdot P(A_1) = 1/4 \times 1/2$

No.

A, B ind $\Rightarrow$ A, B^c ind, A^c, B^c ind

Ind. of n events

$A_1, A_2 \dots A_n$ ind if $\mathbb{P}\left(\bigcap_{i \in I_0} A_i\right) = \prod_{i \in I_0} \mathbb{P}(A_i)$

$\forall I_0 \subseteq \{1, 2 \dots n\}$

Eg 2 bits X, Y. All configs equally likely

$A \to X = 1$

$B \to Y = 1$

$C = X \oplus Y = 1$

Are A, B, C ind.

$\mathbb{P}(A) = \mathbb{P}(B) = \mathbb{P}(C) = \frac{1}{2}$

$P(A \cap B) = \frac{1}{4} \quad [\Sigma_1, 1_3] = \mathbb{P}(A) \cdot \mathbb{P}(B)$

Same for $\mathbb{P}(A \cap C)$, $\mathbb{P}(B \cap C)$.

But $\mathbb{P}(A \cap B \cap C) = 0 \neq \mathbb{P}(A) \cdot \mathbb{P}(B) \cdot \mathbb{P}(C)$

A, B, C not independent